

Lead Scoring Model

X Education — Logistic Regression Case Study

Summary Report

Project	Lead Scoring & Hot Lead Identification
Algorithm	Logistic Regression (L1 pre-selection + p-value + VIF pruning)
Dataset	9,240 leads 37 raw features 17 final features
Goal	Assign Lead Score 0-100; achieve ~80% sensitivity on conversions
AUC-ROC	0.8737 Sensitivity @ threshold 0.30: 83.8%

1. Background & Objective

X Education is an online course provider that markets to working professionals across several digital channels. Despite generating a large volume of leads daily, the company converts only about 30% of them — far below its target. The core challenge is that the sales team spends equal effort on every lead regardless of their actual likelihood to convert, resulting in wasted resources and missed revenue.

The objective of this project was to build a **Logistic Regression model** that assigns a **Lead Score between 0 and 100** to each incoming lead. A higher score indicates a higher probability of conversion ("Hot Lead"), enabling the sales team to prioritise their outreach and achieve the CEO's target of approximately **80% conversion rate** among the leads they choose to contact.

2. Data Overview

The dataset comprised **9,240 historical leads** with 37 attributes covering demographics, behavioural signals (website visits, time on site, page views), lead origin, last activity, occupation, and several binary preference flags. The target variable **Converted** indicated whether a lead resulted in a course enrolment (1) or not (0), with a baseline conversion rate of **38.5%**.

Key observations from EDA:

- Working professionals converted at significantly higher rates than students or unemployed leads.
- Leads originating from the Lead Add Form channel showed the highest conversion rates.
- Total time spent on the website was strongly positively correlated with conversion.
- Data leakage concern:** Lead Quality and Tags were excluded — they are assigned by sales agents after interaction and reflect outcomes rather than lead-level predictors.

3. Data Preparation & Feature Engineering

The raw dataset required substantial cleaning. Columns with more than 35% missing values were removed. The string value "Select" — used when users skipped a form field — was treated as null and replaced with NaN. Numeric columns were imputed with their median (robust to outliers); categorical columns with the mode. Identifier columns (Prospect ID, Lead Number) were dropped as they carry no predictive signal.

Fourteen Yes/No binary columns were mapped to 0/1. High-cardinality categorical features (Lead Source, Last Activity, Specialization, Occupation, City, Last Notable Activity) were capped to their top-7 categories with a residual "Other" bucket to prevent dummy variable explosion. All remaining categorical columns were one-hot encoded with `drop_first=True` to avoid the dummy variable trap.

4. Model Building & Feature Selection

After a 70/30 stratified train-test split and StandardScaler normalisation of numeric features, a three-stage feature selection pipeline was applied to reduce 109 encoded features to a final set of **17 features**:

Stage	Method	In	Out
1	L1-regularised Logistic Regression (C=0.1, liblinear)	109	23
2	Iterative p-value pruning (threshold > 0.05, statsmodels)	23	20
3	VIF pruning (threshold > 5.0) + final p-value check	20	17

The final model was fitted using scikit-learn (C=1.0). Statsmodels coefficients were used for odds ratio and p-value interpretation.

5. Model Performance

0.8737 AUC-ROC (Test)	83.8% Sensitivity @ 0.30	75.4% Specificity @ 0.30	81.0% Accuracy @ 0.50
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The model achieves an AUC-ROC of **0.8737**, indicating strong discriminative power. At the default threshold of 0.50, overall accuracy is 81%. By lowering the classification threshold to **0.30** — calibrated to meet the CEO's ~80% sensitivity target — the model correctly identifies **83.8% of all actual converters** while maintaining specificity at 75.4%, meaning three-quarters of non-converts are correctly excluded from the call list.

6. Top Predictive Features

#	Feature	Odds Ratio	Dir.	Business Meaning
1	Lead Origin: Lead Add Form	30.4x	+	Highest intent signal; actively sought the form.
2	Occupation: Working Professional	15.7x	+	Career-driven; strongest motivation to enrol.
3	Last Notable Activity: SMS Sent	7.4x	+	Recent engagement; follow up immediately.
4	Lead Source: Welingak Website	9.8x	+	High-quality referral source.
5	Total Time Spent on Website	3.0x	+	More browsing = higher product interest.
6	Do Not Email = Yes	0.20x	-	Opted out; far less likely to engage.
7	Lead Profile: Student of SomeSchool	0.05x	-	Converts very rarely; deprioritise.

7. Lead Score Distribution

Each lead was assigned a Lead Score (0-100) equal to the predicted conversion probability multiplied by 100, then bucketed into three tiers:

Tier	Score Range	Count	Actual Conversion Rate
Hot Lead	70 - 100	~2,300	~87%
Warm Lead	40 - 69	~2,100	~52%
Cold Lead	0 - 39	~4,840	~11%

8. Business Recommendations

Normal operations:	Use threshold 0.50. Call all leads with Score ≥ 50 . Gives 81% accuracy with balanced precision and recall.
Intern phase (aggressive):	Lower threshold to 0.30. Call all leads with Score ≥ 30 . Sensitivity rises to 83.8%. Extra intern capacity absorbs the higher false-positive call volume.
Post-target (conservative):	Raise threshold to 0.55-0.60. Call only leads with Score ≥ 55 . Precision increases sharply, minimising wasted calls. Redirect capacity to email nurturing for Warm Leads.
Marketing focus:	Prioritise Lead Add Form as an acquisition channel. Target working professionals in campaigns. Respond to SMS-engaged leads within 24 hours for maximum conversion lift.

9. Learnings & Caveats

This project reinforced several important principles in applied machine learning. First, **data leakage must be identified early** — columns like Tags and Lead Quality appear highly predictive but are assigned post-interaction and would inflate model performance artificially in production. Second, **threshold tuning is as important as model selection**: the sensitivity-specificity curve must be examined before deployment. Third, **feature selection improves interpretability**: reducing 109 features to 17 via three-stage pruning made the model more stable, less prone to overfitting, and far easier to explain to business stakeholders. Finally, missing data with non-random patterns (such as the Asymmetrique indices, missing for ~45% of leads) should be treated with caution — imputing nearly half of a column can distort coefficient estimates.

The model is ready for production deployment. Lead Scores can be computed in real-time as new leads are acquired and fed into the CRM to rank the daily call queue. A quarterly retraining cycle is recommended to account for seasonal drift in lead behaviour.